

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: MindaFinancial

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysian F&B SMEs struggle with thin margins (10–15%) and volatile overheads. While they have data (receipts, POS logs, invoices), they lack the analytical "CFO-level" expertise to turn that data into optimized budgets, accurate forecasts, and waste-reduction strategies.
- **Target Domain:** Financial optimization and automated supply chain management for Malaysian F&B SMEs.
- **Proposed Solution Summary:** MindaFinancial is an AI-native financial partner that ingests unstructured business data (receipts, WhatsApp invoices) and structured POS data to provide **Decision Intelligence**. It shifts the entrepreneur from "Manual Record-Keeper" to "Strategic Decision Maker" by automating inventory forecasting and budget optimization.

2. Background & Business Objective

- **Background of the Problem:** F&B accounting is traditionally reactive rather than proactive. Owners only realize they are losing money at the end of the month. Fragmented data across manual logs and delivery platforms makes it impossible to predict "Rush Hours" or identify "Money Leaks" in real-time.
- **Strategic Importance:** In a USD 16.67 billion market facing 17% ingredient cost hikes, precision is survival. Reducing food waste and optimizing stock reorders directly improves the "Money Cycle" and cash flow for local businesses.
- **Strategic Fit / Impact:** MindaFinancial aligns with the **Malaysian Digital Economy Blueprint**. By integrating the **Z.AI GLM** and offering export paths to **Oracle Cloud**,

AWS S3, and Firebase, it creates a high-performance, secure ecosystem for SMEs to scale their financial operations.

3. Product Purpose

Page / Tab	Key Functionality	Local Context
Dashboard	Real-time financial health & projected cash flow.	Decision Narrative: GLM interprets graphs into human-readable advice.
Stock Management	Traffic light inventory system (Risk Alerts).	AI reorder alerts for Santan, Anchovies, & Oil based on usage.
Rush Prediction	Hourly customer heat maps & prep guides.	Impact of Hari Raya, Monsoon rains, or Stadium concerts .
Vendor Management	Directory with integrated ordering history.	Automated Invoicing: Extraction of data from raw supplier PDFs.
Transactions	Sales history & vendor payment tracking.	Local payment filtering: DuitNow, GrabPay, and Cash .
Menu Optimizer	AI-suggested pricing & margin analysis.	Suggested RM adjustments for Nasi Lemak vs. rising poultry costs.
Waste Tracker	Spoilage logging & sustainability scoring.	Reduction Insights: AI flags "overproduction" patterns to save costs.

3.1. Main Goal of the System: To serve as an **AI-CFO** for F&B entrepreneurs, transforming fragmented, unstructured financial data into a "**Decision Narrative**" that optimizes inventory, reduces waste, and secures profit margins through GLM-driven forecasting.

3.2. Intended Users (Target Audience):

- Operations Managers: Staff responsible for inventory and vendor coordination.
- SME Accountants: Seeking automated data extraction tools to bridge the gap between receipts and ledgers.
- F&B Entrepreneur: Owners of cafes, food trucks, and restaurants needing financial clarity.

4. System Functionalities

4.1. Description: MindaFinancial operates as an AI-powered financial decision engine for Malaysian F&B SMEs. Using a multimodal data ingestion pipeline, the system closes the gap between raw operational activity and financial clarity, giving business owners the kind of real-time cost visibility and forward-looking insight that was previously only viable for enterprises with dedicated finance teams. The result is a shift from reactive, end-of-month accounting to proactive, day-to-day financial control.

4.2. Key Functionalities:

- **Multimodal Data Ingestion:** Accepts receipts, Whatsapp invoice images, supplier PDFs and POS exports, from which the GLM will auto-extract vendor details, quantities and amounts, thus eliminating manual data entry.
- **Decision Narrative Engine:** Synthesises data across sales, vendor spend, and stock levels into plain-language financial advice, flagging the biggest cost risks and margin opportunities daily, without requiring the owner to interpret a single chart.
- **Predictive Rush & Inventory Forecasting:** Cross-references historical POS patterns with local demand signals such as public holidays, monsoon season, and

nearby events to project ingredient consumption and flag restocking needs before they become a problem.

- **Automated Vendor & Invoice Management:** Maintains a live vendor directory populated directly from extracted invoice data, tracking order history and flagging price anomalies to give owners full visibility over procurement spend.
- **Menu Margin Optimizer:** Monitors ingredient costs against menu pricing in real time, flagging when a dish's margin slips below where it should be, fit with price adjustment recommendations.
- **Waste & Spoilage Intelligence:** Detects overproduction patterns by comparing purchase volumes against actual sales, quantifying the financial cost of recurring waste and recommending targeted reductions to recover lost margin.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: MindaFinancial is powered by Z.AI GLM. Its strong Malay/English multilingual capability is essential for processing mixed-language Malaysian supplier documents, while its long-context handling supports multi-page invoice PDFs without degradation. Z.AI GLM's reliable structured JSON output makes it well-suited for translating unstructured receipts into ledger-ready data, and its cost-efficient inference profile aligns with the high-frequency, lightweight query patterns of SME-scale usage.

4.3.2. Prompting Strategy: As MindaFinancial uses a multi-step agentic prompting approach, a single prompt cannot reliably carry a raw invoice image all the way through to a financial recommendation. Therefore, the team broke down the pipeline into three sequential stages (extraction, analysis, and narrative), each with its own targeted prompt. This staged approach reduces hallucination risk, allows output validation between steps, and produces more consistent, trustworthy financial advice than a single end-to-end prompt would.

4.3.3. Context & Input Handling: The system accepts supplier PDFs up to 20 pages, WhatsApp and receipt images up to 5MB, and POS exports up to 10,000 rows. Inputs exceeding these limits are handled through progressive chunking, where they are split into logical segments, processed independently, and reassembled into a unified output. Duplicate entries across chunks are deduplicated before any data is committed. Corrupted or fully illegible files are rejected outright, with a plain user-facing prompt to re-upload a clearer version.

4.3.4. Fallback & Failure Behavior: When the GLM returns an incomplete or structurally invalid response, the system automatically retries with a more constrained prompt variant. Extractions that fall below a confidence threshold are flagged with a "Please verify" warning and held for user confirmation before updating any financial records, preventing silent errors from entering the ledger. Repeated failures on the same document escalate to a Manual Review Queue, where the owner or accountant is notified via dashboard to intervene. All failed events are logged to continuously improve prompt performance over time.

5. User Stories & Use Cases

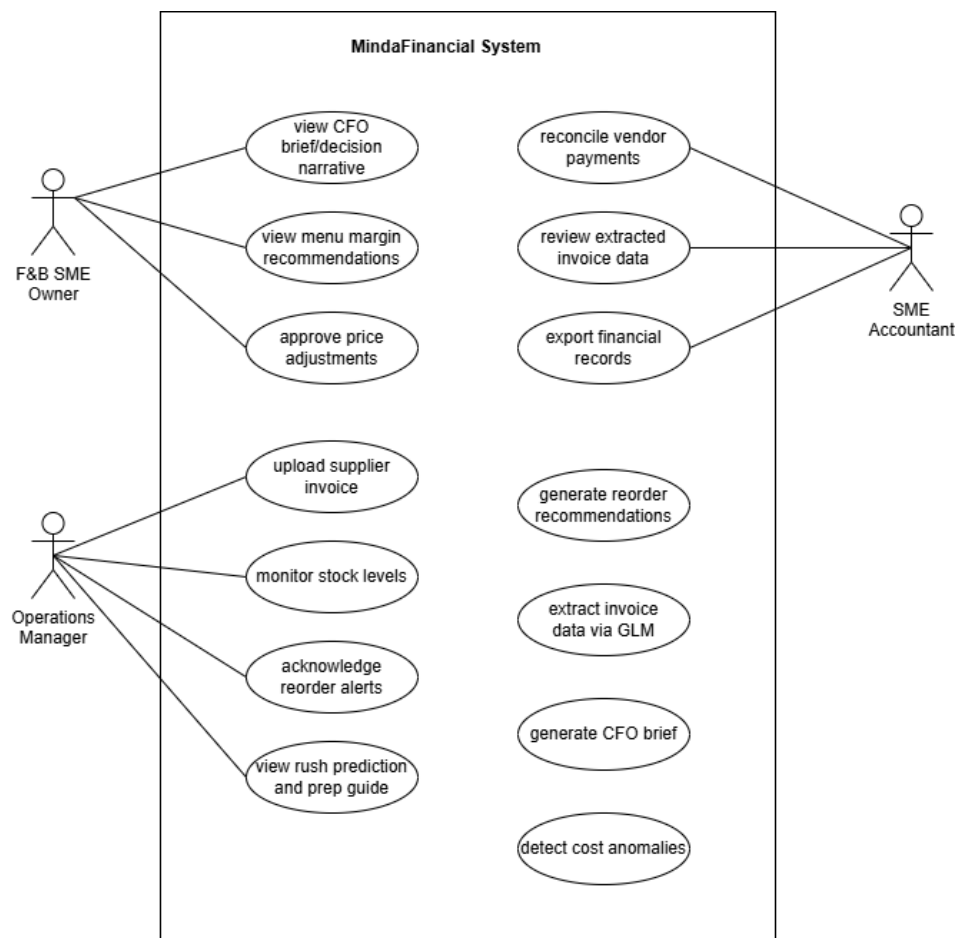
- **User Stories:**

- *"As an F&B SME owner, I want supplier invoices processed automatically upon receipt, so that my financial records are always up to date without manual effort."*
- *"As an operations manager, I want stock risk alerts tied to projected financials, so that I can prioritise reorders based on cost impact rather than guesswork."*
- *"As an SME accountant, I want vendor payments automatically matched against extracted invoice data, so that month-end reconciliation isn't so time-consuming."*
- *"As an F&B SME owner, I want dish-level margin alerts when ingredient costs rise, so that I can adjust pricing before profits are eroded."*

- **Use Cases (Main Interactions):**

- **Invoice-to-Ledger Flow:** The owner forwards a supplier invoice, the GLM extracts the vendor details, line items, and prices, and upon confirmation, the ledger, vendor directory, and stock levels all update simultaneously.

- **Financial Risk Alert Flow:** The system monitors spending and stock against budget thresholds, and when an anomaly is detected, the dashboard surfaces a plain-language alert with the cost impact and recommended action.
- **CFO Brief Generation Flow:** At the end of each day or week, the GLM synthesises sales, vendor spend, waste, and margin performance into a prioritised summary of the top financial risks and opportunities, with direct links into the relevant modules for immediate action.
- **Menu Margin Optimization Flow:** When an ingredient cost crosses a defined threshold, the Menu Optimizer recomputes the margins of all affected dishes and surfaces a price adjustment recommendation for the owner to review.
- **Use Case Diagram:**



6. Features Included (Scope Definition)

- Multimodal invoice ingestion from WhatsApp images, supplier PDFs, and scanned receipts, auto-extracted into structured financial records.
- Daily plain-language CFO Briefs in Bahasa Malaysia and English, accessible to non-finance-trained owners.
- Live dish-level margin tracking with RM price adjustment recommendations that move with ingredient costs.
- Traffic-light inventory monitoring with reorder alerts for Malaysian F&B staples including santan, anchovies, and cooking oil.
- Rush Prediction module linking demand forecasts directly to procurement planning and cost projections.
- Vendor directory with automated invoice reconciliation, order history, and price anomaly detection.
- Waste and overproduction tracking with ringgit-value cost impact and purchase reduction recommendations.

7. Features Not Included (Scope Control)

- **Automated Vendor Ordering:** The "Approve Order" button triggers a UI success state and local inventory update, but it does not send real-world emails or API calls to actual suppliers like Fresh Farm Supplies.
- **Multi-tenant User Authentication:** The current version is a single-session demo designed for one restaurant manager and does not support separate login credentials for multiple staff members.
- **Historical Data Analytics & Trend Reporting:** Current AI insights are generated based on present inventory levels and immediate usage patterns. Long-term historical data persistence including year-over-year growth charts, seasonal trend exports, and full PDF audit logs are not implemented in this version.

8. Assumptions & Constraints

- **LLM Cost Constraint:** To keep inference costs low, we utilize Data Summarization. Instead of sending thousands of raw sales rows to the AI, we pre-process totals on the frontend. This keeps our average token usage below 2,000 tokens per analysis session.
- **Technical Constraints:** The dashboard is optimized for Desktop/Tablet viewports to ensure business managers can read complex data charts. Mobile responsiveness is limited to basic viewing.
- **Performance Constraints:** To maintain high speed, the AI-generated "Rationales" are capped at 150 words per recommendation.
- **User Input:** The AI only recommends stock adjustments and it cannot execute them without a human clicking "Approve."

9. Risks & Questions Throughout Development

- **Design Similarity:** How do we make the design look unique instead of a generic AI template?
While we used an AI Figma maker for initial scaffolding, we manually overrode the design with a custom Malaysian business feel. We chose a specific "Success Green" color theme and added local details, like alerts for Hari Raya and local vendor names to ensure the product feels unique to the local market.
- **Frame Stability:** What is the risk of alignment errors when converting AI designs to functional code?
Exported AI frames often fail to be responsive. We mitigated this by manually rebuilding the navigation and grid systems in Tailwind CSS, ensuring that critical data remains visible and stable regardless of the user's screen resolution.
- **Drawing Interpretation:** How do we know the charts are showing the right numbers and aren't just drawings?
AI design tools usually just draw a chart image that never changes. We replaced those fake drawings with Recharts logic. This means our charts are alive and they actually read

the numbers from our inventory list and change automatically, so the business owner can trust what they see.